

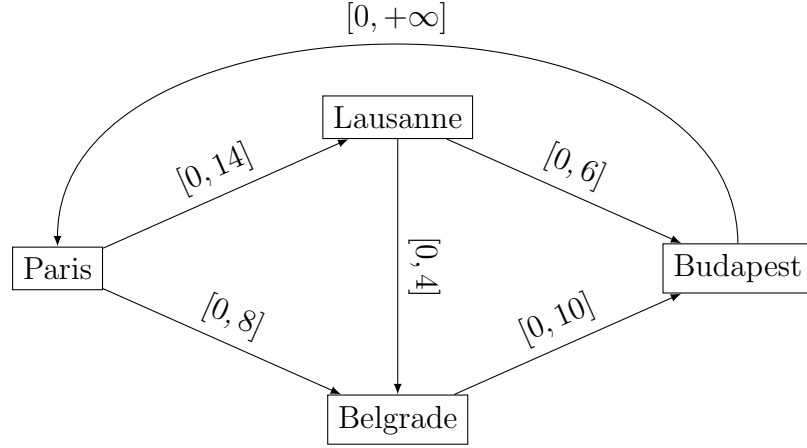
Modeling: maximum flow

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Solution of the practice quiz

1. The maximum flow problem has no cost associated with the arcs. To model the maximum flow problem as a transshipment problem, we need to include a dummy arc from Budapest to Paris, associated with a negative cost (-1). We also impose a circulation, which means that we impose that each node is a transit node, with a divergence of zero. Therefore, each unit of flow reaching Budapest is sent back to Paris through the dummy arc. As the real arcs have zero costs, by assigning a cost of -1 to the dummy arc, the opposite of the total cost represents the number of units of flow that are sent back from Budapest to Paris through the dummy arc. Minimizing the total cost is therefore equivalent to maximizing the total flow through the network.

The lower capacity of each real arc is zero, so that the arcs cannot be used in the reverse direction. The upper capacity corresponds to the maximum number of train for one day between the corresponding cities. The artificial arc is also bounded below by zero, but is not bounded above. We use the notation $[\ell_{ij}, u_{ij}]$ to denote the capacity bounds on the network.



A transshipment problem is written

$$\min_{x \in \mathbb{R}^n} \sum_{(i,j) \in \mathcal{A}} c_{ij} x_{ij}.$$

subject to the flow conservation constraints,

$$\text{div}(x)_i = s_i \quad \forall i \in \mathcal{N},$$

and the bounds on the flows:

$$\ell_{ij} \leq x_{ij} \leq u_{ij}, \quad \forall (i, j) \in \mathcal{A}.$$

In our case,

- $\mathcal{N}$ is the set of nodes: *Budapest* (*Bu*), *Paris* (*P*), *Lausanne* (*L*) and *Belgrade* (*Be*).
- $\mathcal{A}^+ = \mathcal{A} \cup \{(Bu, P)\}$ is the set combining the original arcs and the dummy arc.
- Costs: $c_{ij} = 0$, $\forall (i, j) \in \mathcal{A}$ and $c_{Bu, P} = -1$.
- Supply: as we impose a circulation, the divergence at each node must be zero: $s_i = 0$, $\forall i \in \mathcal{N}$.
- Bounds: $0 \leq x_{ij} \leq u_{ij}$, $\forall (i, j) \in \mathcal{A}$, and $x_{Bu, P} \geq 0$.

The transshipment problem is

$$\min_x -x_{\text{Bu,P}}.$$

subject to

$$\begin{aligned} \sum_{j|(i,j) \in \mathcal{A}^+} x_{ij} - \sum_{k|(k,i) \in \mathcal{A}^+} x_{ki} &= 0, \quad \forall i \in \mathcal{N} \\ x_{ij} &\leq u_{ij}, \quad \forall (i,j) \in \mathcal{A} \\ x_{ij} &\geq 0, \quad \forall (i,j) \in \mathcal{A}^+. \end{aligned}$$

In our case, it is

$$\min_x -x_{\text{Bu,P}}.$$

subject to

$$\begin{aligned} x_{\text{P,L}} + x_{\text{P,Be}} - x_{\text{Bu,P}} &= 0, \\ -x_{\text{P,L}} + x_{\text{L,Be}} + x_{\text{L,Bu}} &= 0, \\ -x_{\text{P,Be}} - x_{\text{L,Be}} + x_{\text{Be,Bu}} &= 0, \\ -x_{\text{L,Bu}} - x_{\text{Be,Bu}} + x_{\text{Bu,P}} &= 0, \\ x_{\text{P,L}} &\leq 14, \\ x_{\text{P,Be}} &\leq 8, \\ x_{\text{L,Be}} &\leq 4, \\ x_{\text{L,Bu}} &\leq 6, \\ x_{\text{Be,Bu}} &\leq 10, \\ x_{\text{P,L}}, x_{\text{P,Be}}, x_{\text{L,Be}}, x_{\text{L,Bu}}, x_{\text{Be,Bu}}, x_{\text{Bu,P}} &\geq 0. \end{aligned}$$

2. In order to solve the problem using the simplex algorithm, we start by writing it in standard form by introducing the slack variables x^s .

$$\min_x -x_{\text{Bu,P}}.$$

subject to

$$\begin{array}{rcl}
x_{P,L} & & +x_{P,L}^s = 14 \\
& x_{P,Be} & +x_{P,Be}^s = 8 \\
& & x_{L,Be} & +x_{L,Be}^s = 4 \\
& & & x_{L,Bu} & +x_{L,Bu}^s = 6 \\
& & & & x_{Be,Bu} & +x_{Be,Bu}^s = 10 \\
x_{P,L} & +x_{P,Be} & & & & -x_{Bu,P} & = 0 \\
-x_{P,L} & & +x_{L,Be} & +x_{L,Bu} & & & = 0 \\
& -x_{P,Be} & -x_{L,Be} & & +x_{Be,Bu} & & = 0 \\
& & & -x_{L,Bu} & -x_{Be,Bu} & +x_{Bu,P} & = 0 \\
x_{P,L}, & x_{P,Be}, & x_{L,Be}, & x_{L,Bu}, & x_{Be,Bu}, & x_{Bu,P} & \geq 0, \\
x_{P,L}^s, & x_{P,Be}^s, & x_{L,Be}^s, & x_{L,Bu}^s, & x_{Be,Bu}^s & & \geq 0,
\end{array}$$

It corresponds to

$$A = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 1 \\ 1 & 1 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 \\ -1 & 0 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & -1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 & -1 & 1 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}, \quad b = \begin{pmatrix} 14 \\ 8 \\ 4 \\ 6 \\ 10 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}.$$

The vector of costs is:

$$c = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ -1 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

An auxiliary variable y_i is associated with each of the equality constraints in order to obtain the auxiliary problem of Phase I. The iterations of Phase I are reported below.

Iteration 0

$x_{P,L}$	$x_{P,Be}$	$x_{L,Be}$	$x_{L,Bu}$	$x_{Be,Bu}$	$x_{Bu,P}$	$x_{P,L}^s$	$x_{P,Be}^s$	$x_{L,Be}^s$	$x_{L,Bu}^s$	$x_{Be,Bu}^s$	y_1	y_2	y_3	y_4	y_5	y_6	y_7	y_8	y_9	
1	0	0	0	0	0	1	0	0	0	0	1	0	0	0	0	0	0	0	0	14
0	1	0	0	0	0	0	1	0	0	0	0	1	0	0	0	0	0	0	0	8
0	0	1	0	0	0	0	0	1	0	0	0	0	1	0	0	0	0	0	0	4
0	0	0	1	0	0	0	0	0	1	0	0	0	0	1	0	0	0	0	0	6
0	0	0	0	1	0	0	0	0	0	1	0	0	0	0	1	0	0	0	0	10
1	1	0	0	0	-1	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0
-1	0	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0
0	-1	-1	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0
0	0	0	-1	-1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0
-1	-1	-1	-1	-1	-0	-1	-1	-1	-1	-1	0	0	0	0	0	0	0	0	0	-42

Iteration 1

$x_{P,L}$	$x_{P,Be}$	$x_{L,Be}$	$x_{L,Bu}$	$x_{Be,Bu}$	$x_{Bu,P}$	$x_{P,L}^s$	$x_{P,Be}^s$	$x_{L,Be}^s$	$x_{L,Bu}^s$	$x_{Be,Bu}^s$	y_1	y_2	y_3	y_4	y_5	y_6	y_7	y_8	y_9	
0	-1	0	0	0	1	1	0	0	0	0	1	0	0	0	0	-1	0	0	0	14
0	1	0	0	0	0	0	1	0	0	0	0	1	0	0	0	0	0	0	0	8
0	0	1	0	0	0	0	0	1	0	0	0	0	1	0	0	0	0	0	0	4
0	0	0	1	0	0	0	0	0	1	0	0	0	0	1	0	0	0	0	0	6
0	0	0	0	1	0	0	0	0	0	1	0	0	0	0	1	0	0	0	0	10
1	1	0	0	0	-1	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0
0	1	1	1	0	-1	0	0	0	0	0	0	0	0	0	0	1	1	0	0	0
0	-1	-1	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0
0	0	0	-1	-1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0
0	0	-1	-1	-1	-1	-1	-1	-1	-1	-1	0	0	0	0	0	1	0	0	0	-42

Iteration 2

$x_{P,L}$	$x_{P,Be}$	$x_{L,Be}$	$x_{L,Bu}$	$x_{Be,Bu}$	$x_{Bu,P}$	$x_{P,L}^s$	$x_{P,Be}^s$	$x_{L,Be}^s$	$x_{L,Bu}^s$	$x_{Be,Bu}^s$	y_1	y_2	y_3	y_4	y_5	y_6	y_7	y_8	y_9	
0	-1	0	0	0	1	1	0	0	0	0	1	0	0	0	0	-1	0	0	0	14
0	1	0	0	0	0	0	1	0	0	0	0	1	0	0	0	0	0	0	0	8
0	-1	0	-1	0	1	0	0	1	0	0	0	0	1	0	0	-1	-1	0	0	4
0	0	0	1	0	0	0	0	0	1	0	0	0	0	1	0	0	0	0	0	6
0	0	0	0	1	0	0	0	0	0	1	0	0	0	0	1	0	0	0	0	10
1	1	0	0	0	-1	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0
0	1	1	1	0	-1	0	0	0	0	0	0	0	0	0	0	1	1	0	0	0
0	0	0	1	1	-1	0	0	0	0	0	0	0	0	0	0	1	1	1	0	0
0	0	0	-1	-1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0
0	1	0	0	-1	-2	-1	-1	-1	-1	-1	0	0	0	0	0	2	1	0	0	-42

Iteration 3

$x_{P,L}$	$x_{P,Be}$	$x_{L,Be}$	$x_{L,Bu}$	$x_{Be,Bu}$	$x_{Bu,P}$	$x_{P,L}^s$	$x_{P,Be}^s$	$x_{L,Be}^s$	$x_{L,Bu}^s$	$x_{Be,Bu}^s$	y_1	y_2	y_3	y_4	y_5	y_6	y_7	y_8	y_9
0	-1	0	0	0	1	1	0	0	0	0	1	0	0	0	0	-1	0	0	0
0	1	0	0	0	0	0	1	0	0	0	0	1	0	0	0	0	0	0	0
0	-1	0	-1	0	1	0	0	1	0	0	0	0	1	0	0	-1	-1	0	0
0	0	0	1	0	0	0	0	0	1	0	0	0	0	1	0	0	0	0	0
0	0	0	-1	0	1	0	0	0	0	1	0	0	0	0	1	-1	-1	0	0
1	1	0	0	0	-1	0	0	0	0	0	0	0	0	0	0	1	0	0	0
0	1	1	1	0	-1	0	0	0	0	0	0	0	0	0	0	1	1	0	0
0	0	0	1	1	-1	0	0	0	0	0	0	0	0	0	0	1	1	1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	1
0	1	0	1	0	-3	-1	-1	-1	-1	-1	0	0	0	0	0	3	2	1	0

Iteration 4

$x_{P,L}$	$x_{P,Be}$	$x_{L,Be}$	$x_{L,Bu}$	$x_{Be,Bu}$	$x_{Bu,P}$	$x_{P,L}^s$	$x_{P,Be}^s$	$x_{L,Be}^s$	$x_{L,Bu}^s$	$x_{Be,Bu}^s$	y_1	y_2	y_3	y_4	y_5	y_6	y_7	y_8	y_9
0	0	0	1	0	0	1	0	-1	0	0	1	0	-1	0	0	0	1	0	0
0	1	0	0	0	0	0	1	0	0	0	0	1	0	0	0	0	0	0	0
0	-1	0	-1	0	1	0	0	1	0	0	0	0	1	0	0	-1	-1	0	0
0	0	0	1	0	0	0	0	0	1	0	0	0	0	1	0	0	0	0	0
0	1	0	0	0	0	0	0	-1	0	1	0	0	-1	0	1	0	0	-1	0
1	0	0	-1	0	0	0	0	1	0	0	0	0	1	0	0	0	-1	0	0
0	0	1	0	0	0	0	0	1	0	0	0	0	1	0	0	0	0	0	0
0	-1	0	0	1	0	0	0	1	0	0	0	0	1	0	0	0	0	1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	1
0	-2	0	-2	0	0	-1	-1	2	-1	-1	0	0	3	0	0	0	-1	1	0

Iteration 5

$x_{P,L}$	$x_{P,Be}$	$x_{L,Be}$	$x_{L,Bu}$	$x_{Be,Bu}$	$x_{Bu,P}$	$x_{P,L}^s$	$x_{P,Be}^s$	$x_{L,Be}^s$	$x_{L,Bu}^s$	$x_{Be,Bu}^s$	y_1	y_2	y_3	y_4	y_5	y_6	y_7	y_8	y_9
0	0	0	1	0	0	1	0	-1	0	0	1	0	-1	0	0	0	1	0	0
0	0	0	0	0	0	0	1	1	0	-1	0	1	1	0	-1	0	0	1	0
0	0	0	-1	0	1	0	0	0	0	1	0	0	0	0	1	-1	-1	0	0
0	0	0	1	0	0	0	0	0	1	0	0	0	0	1	0	0	0	0	0
0	1	0	0	0	0	0	0	-1	0	1	0	0	-1	0	1	0	0	-1	0
1	0	0	-1	0	0	0	0	1	0	0	0	0	1	0	0	0	-1	0	0
0	0	1	0	0	0	0	0	1	0	0	0	0	1	0	0	0	0	0	0
0	0	0	0	1	0	0	0	0	0	1	0	0	0	0	1	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	1
0	0	0	-2	0	0	-1	-1	0	-1	1	0	0	1	0	2	0	-1	-1	0

Iteration 6

$x_{P,L}$	$x_{P,Be}$	$x_{L,Be}$	$x_{L,Bu}$	$x_{Be,Bu}$	$x_{Bu,P}$	$x_{P,L}^s$	$x_{P,Be}^s$	$x_{L,Be}^s$	$x_{L,Bu}^s$	$x_{Be,Bu}^s$	y_1	y_2	y_3	y_4	y_5	y_6	y_7	y_8	y_9
0	0	0	0	0	0	1	0	-1	-1	0	1	0	-1	-1	0	0	1	0	0
0	0	0	0	0	0	0	1	1	0	-1	0	1	1	0	-1	0	0	1	0
0	0	0	0	0	1	0	0	0	1	1	0	0	0	1	1	-1	-1	-1	0
0	0	0	1	0	0	0	0	0	1	0	0	0	0	1	0	0	0	0	0
0	1	0	0	0	0	0	0	-1	0	1	0	0	-1	0	1	0	0	-1	0
1	0	0	0	0	0	0	0	1	1	0	0	0	1	1	0	0	-1	0	0
0	0	1	0	0	0	0	0	1	0	0	0	0	1	0	0	0	0	0	0
0	0	0	0	1	0	0	0	0	0	1	0	0	0	0	1	0	0	0	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	1
0	0	0	0	0	0	-1	-1	0	1	1	0	0	1	2	2	0	-1	-1	0

Iteration 7

$x_{P,L}$	$x_{P,Be}$	$x_{L,Be}$	$x_{L,Bu}$	$x_{Be,Bu}$	$x_{Bu,P}$	$x_{P,L}^S$	$x_{P,Be}^S$	$x_{L,Be}^S$	$x_{L,Bu}^S$	$x_{Be,Bu}^S$	y_1	y_2	y_3	y_4	y_5	y_6	y_7	y_8	y_9	
0	0	0	0	0	0	1	0	-1	-1	0	1	0	-1	-1	0	0	1	0	0	4
0	0	0	0	0	0	0	1	1	0	-1	0	1	1	0	-1	0	0	1	0	2
0	0	0	0	0	1	0	0	0	1	1	0	0	0	1	1	-1	-1	-1	0	16
0	0	0	1	0	0	0	0	0	1	0	0	0	0	1	0	0	0	0	0	6
0	1	0	0	0	0	0	0	-1	0	1	0	0	-1	0	1	0	0	-1	0	6
1	0	0	0	0	0	0	0	1	1	0	0	0	1	1	0	0	-1	0	0	10
0	0	1	0	0	0	0	0	1	0	0	0	0	1	0	0	0	0	0	0	4
0	0	0	0	1	0	0	0	0	0	1	0	0	0	0	1	0	0	0	0	10
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	1	0
0	0	0	0	0	0	0	-1	-1	0	1	1	0	0	1	2	0	0	-1	0	-2

Iteration 8

$x_{P,L}$	$x_{P,Be}$	$x_{L,Be}$	$x_{L,Bu}$	$x_{Be,Bu}$	$x_{Bu,P}$	$x_{P,L}^S$	$x_{P,Be}^S$	$x_{L,Be}^S$	$x_{L,Bu}^S$	$x_{Be,Bu}^S$	y_1	y_2	y_3	y_4	y_5	y_6	y_7	y_8	y_9	
0	0	0	0	0	0	1	0	-1	-1	0	1	0	-1	-1	0	0	1	0	0	4
0	0	0	0	0	0	0	1	1	0	-1	0	1	1	0	-1	0	0	1	0	2
0	0	0	0	0	1	0	0	0	1	1	0	0	0	1	1	-1	-1	-1	0	16
0	0	0	1	0	0	0	0	0	1	0	0	0	0	1	0	0	0	0	0	6
0	1	0	0	0	0	0	0	-1	0	1	0	0	-1	0	1	0	0	-1	0	6
1	0	0	0	0	0	0	0	1	1	0	0	0	1	1	0	0	-1	0	0	10
0	0	1	0	0	0	0	0	1	0	0	0	0	1	0	0	0	0	0	0	4
0	0	0	0	1	0	0	0	0	0	1	0	0	0	0	1	0	0	0	0	10
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	1	0
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	1	0

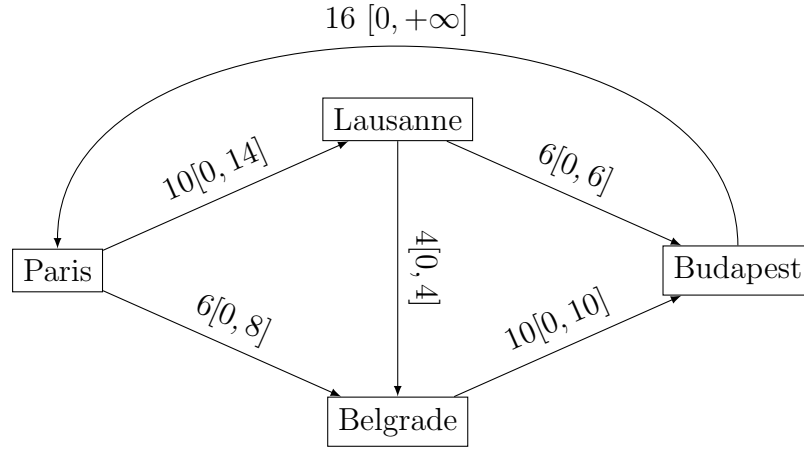
The optimal value is 0. Therefore, we have found a feasible solution of the original problem. But the auxiliary variable y_9 is still in the basis. It is not possible to pivot it out, as all potential pivots are zero. Actually, the transshipment problem as we have formulated it, has always one redundant constraint. As a result, the auxiliary variable stays in the basis. The corresponding constraint can be removed. The columns of the auxiliary variables are then removed, and the reduced costs calculated, to obtain the first tableau for Phase II:

[illegible]

It is actually optimal, as all reduced costs are non negative. We end up with the optimal solution:

$$\begin{aligned}x_{P,L} &= 10 \\x_{P,Be} &= 6 \\x_{L,Bu} &= 6 \\x_{L,Be} &= 4 \\x_{Be,Bu} &= 10.\end{aligned}$$

In the following picture, the optimal flow is represented on each arc, next to the bounds.



Note that all the values are integers, although not explicitly required. The optimal value is -16 , which corresponds to the total flow from Paris to Budapest.

Note also that it's not the only optimal solution. Indeed, in the last tableau, the non basic variable $x_{L,Be}^s$ has a reduced cost equal to 0. Therefore, we can move to another optimal tableau by introducing $x_{L,Be}^s$ in the basis instead of $x_{P,Be}^s$. We obtain:

$$\begin{aligned}x_{P,L} &= 8 \\x_{P,Be} &= 8 \\x_{L,Bu} &= 6 \\x_{L,Be} &= 2 \\x_{Be,Bu} &= 10.\end{aligned}$$

It is represented by the following picture.

